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Temperature Impact on Perovskite Solar Cells Under Operation

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Abstract: Perovskite solar cells (PSC) have emerged as a promising substitute of conventional silicon panels showing the fastest power conversion efficiency evolution within photovoltaic field, from 3.8 % to 23.7 % in few years. However, PSC thermal stability is still an obstacle to their commercialization and so the temperature effect on the mesoporous triple cation perovskite solar cells with two different hole extraction materials (spiro-MeOTAD and PTAA) was assessed in this work. The cells were exposed to thermal stress between -5 °C and 80 °C and their photovoltaic performance was monitored in-situ to reproduce real operating conditions. At low temperatures, devices presented very stable values (average loss < 5 %), whilst as temperature increases a significant decrease in the open circuit potential and short-circuit current was observed. X-ray diffraction showed no modification of the perovskite crystal structure with temperature. On the other hand, electron scanning microscopy and X-ray photoelectron spectroscopy helped to conclude that temperature had a great impact in the hole extraction layer. The cell performance loss was attributed to the evaporation of additives added to the hole extraction layer to enhance its conductivity. Although the decrease in power conversion efficiency at 80 °C is slightly higher for PTAA cells, spiro-MeOTAD cells presented a higher irreversible loss of (21.6 ± 2.3) % after thermal-stress tests while PTAA devices showed only a loss of (8.2 ± 1.6) %.

Introduction

In less than a decade the efficiency of perovskite solar cells (PSC) raised from 3.8 % to 23.7 %.^[1] This fast improvement in terms of power conversion efficiency (PCE) was mainly possible by tailoring the formation and composition of perovskite materials,^[2-5] new deposition methods^[6-8] and the use of new hole and electron extraction layers.^[9-14] Although perovskites are very efficient as light absorbers and easy to produce, complete devices still presenting some stability challenges that is hampering their large-scale production. Perovskite solar cells are very sensitive to temperature, moisture and UV light and, for that reason, understanding the effect of these environmental elements in devices under operation is of crucial importance. Temperature is particularly critical since normal operating environment for photovoltaic devices is under direct sunlight, the cell temperature can be up to 45 °C higher than the environmental temperature.^[15] So far, temperature dependent studies on PSCs have been very limited compared to the numerous studies on improving their power output at room temperature.

Due to the unprecedented photovoltaic properties of the organometal halide perovskite crystals, the majority of studies focused on the chemical stability of the perovskite material as a function of the temperature. Actually, it is well known that the crystal structure is significantly affected by temperature and that perovskite crystals show three phases, namely α , β , and γ phases,

depending on the crystallization temperature. According with some authors,^[16-18] in $\text{CH}_3\text{NH}_3\text{PbI}_3$ perovskites at 100 °C an intermediate phase of the perovskite absorber, between tetragonal and cubic phase, starts appearing and the PSC device loses performance. Also, when perovskite is exposed for a long time to 85 °C it starts decomposing into PbI_2 .^[18] Several research groups progressed on improving the thermal stability of PSCs by developing more stable hybrid perovskites, and the addition of small amounts of inorganic cesium and rubidium cations resulted in highly monolithic grains of more pure perovskite.^[2-3] In fact, it has been observed that the “cation cascade” perovskite is quite stable under dry conditions and at temperatures up to 85 °C, showing apparently no change in color neither in the X-ray diffraction (XRD) patterns of the devices submitted to 85 °C. Following the attempt to overcome thermal instability, different architectures and deposition techniques^[2,7,12,19] have been studied, which led to verify the additional thermal instability of hole extraction materials. In the best performing devices spiro-MeOTAD [2,2',7,7'-tetrakis(N,N-di-p-methoxyphenyl-amine)9,9'-spirobifluorene] is used as hole extraction layer (HEL). Although the glass transition temperature of spiro-MeOTAD is around 125 °C,^[20] the use of dopants like Li-bis(trifluoromethanesulfonyl) imide (Li-TFSI) and tert-butylpyridine (TBP) to enhance electrical conductivity actually reduces this temperature to values below 100 °C. So, for these devices, temperature is a key factor since spiro-MeOTAD with additives crystallizes when subjected to thermal-stress, responsible for reducing the hole mobility and electrical contact of perovskite/HEL interface.^[21] Besides, the interface between perovskite and HEL may be also affected promoting the degradation of HEL due to the I^- migration. According to Kim *et al.*, at elevated temperatures iodide ions from the perovskite layer diffuse into the spiro-MeOTAD and act as reducing agents decreasing its oxidation and, consequently, its conductivity properties.^[17] Moreover, Jena *et al.*^[22] attributed the PSC performance degradation under temperature stress to crystallization and photo-oxidation of spiro-MeOTAD; they observed that when overheating a perovskite film, free of spiro-MeOTAD, to promote its degradation to PbI_2 , the PSC device did not show significant performance losses. This led to the conclusion that the textural modification of the spiro-MeOTAD layer (generation of pinholes at high temperature) and certain unidentified modifications at the perovskite/spiro-MeOTAD interface seem to be the main reasons for the performance drop of the cells at high temperature. Zheng *et al.*^[23] also ascribed the

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loss of performance of formamidinium perovskite devices to the loss of dopants from the HEL. These authors observed that when using a dopant-free HEL (P3HT – poly(3-hexylthiophene-2,5-diyl)) the devices retained the initial efficiency and showed no apparent performance change after 800 h storage in the temperature range of 20 - 85 °C. More recently, Magomedov *et al.*^[24] showed that oxidized HEL species can interact with *tert*-butylpyridine additive forming pyridinated products that influence negatively the performance of the device, even in short aging times. Moreover, Kasparavicius *et al.*^[25] also showed that oxidized HEL starts to degrade at 100 °C and part reverts back to the original unoxidized material, and other part reacts with TBP. Additionally, Li-TFSI and FK209, also responsible for the oxidation of HELs, do not affect the degradation rate. Thus, the authors concluded that even if the dopants are important to oxidize the HELs to increase their conductivities and consequently to obtain higher power conversion efficiencies, the instability of the oxidized HELs can be actually responsible for the low-performing spiro-MeOTAD based-devices at elevated temperatures. Besides, high temperatures also proved to induce gold migration to inner-layers. According to Domanski *et al.* a temperature of 70 °C is enough to induce gold migration from the back contact through the HEL (spiro-MeOTAD) and into perovskite layer affecting consequently the performance of the cell.^[26]

Important to mention that the aforementioned temperature tests were not made under real operation conditions; the cells were exposed to thermal-stress and the characteristic curves obtained at room temperature. In outdoor conditions, the PSC device produces electricity at the operating temperature conditions. Moreover, most of the studies concerning temperature stress were performed under nitrogen atmosphere or with cells sealed under inert atmosphere, what will be very difficult to realize in large-scale industrialization, not only in terms of costs but also in terms of equipment limitations. The present work aims at mimicking as close as possible real operating conditions. PSC devices sealed under room conditions were submitted to thermal-stress between -5 °C to 80 °C and characterized at the stress temperature. Mesoporous triple-cation perovskite cells using a spiro-MeOTAD or PTAA HEL were prepared and then sealed under ambient conditions of oxygen and humidity. The performance degradation of the devices was investigated by photocurrent-potential (*I*-*V*), X-ray diffraction (XRD), scanning electron microscopy (SEM) and X-ray photoelectron spectroscopy (XPS).

Results and Discussion

Thermal-stress measurements were conducted in perovskite solar cells prepared from different batches and characterized in different days. The experiments performed at temperatures below room temperature (12 °C, 7 °C, 0 °C and -5 °C) were conducted first; then, these cells were stabilized at 22 °C and the correspondent photovoltaic performance assessed again. The same batch of 10 cells was used at all low-temperature experiments. The cells presented very stable results for all

photovoltaic parameters with an average decrease of approximately 5 % when tested at -5 °C compared with the initial PCE at 22 °C. This loss of efficiency was mainly due to a decrease in the V_{oc} (open circuit voltage) and FF (fill factor) and consequently an increase of the series resistance. Nevertheless, the observed PCE decrease at -5 °C was then fully reversible when devices are characterized again at room-temperature - Figure 1. Spiro-MeOTAD mobility has been characterized at different temperatures by Rana *et al.* and it was observed that as temperature decreases, the charge mobility also decreases rapidly suggesting a thermally activated transport mechanism.^[27] This type of transport process has been observed in organic semiconductors by many authors.^[28-31]

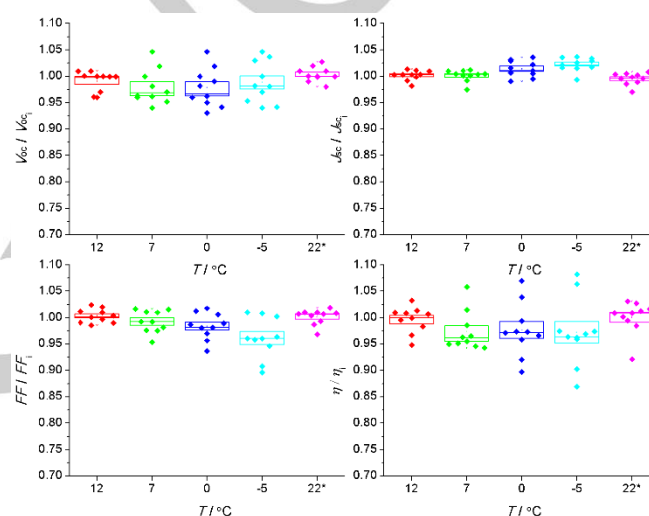


Figure 1. Photovoltaic performance of complete PSC devices at temperatures between -5 °C and 12 °C. The photovoltaic performance values were normalized by the initial PCE obtained at 22 °C for each cell. It was also added the normalized performance parameters of cells characterized at 22 °C after low-temperature stress tests (22*).

Since no significant changes in the performance were observed, the same cell batch was exposed to higher temperatures, stepwise from 30 °C up to 80 °C. As temperature increases, a very significant decrease in the V_{oc} and J_{sc} (short-circuit current) was observed, leading to a decrease in the overall performance of the devices – Figure 2. When the devices were tested at 80 °C the PCE diminished in average of (36.0 ± 5.5) % compared with the initial PCE of each cell at 22 °C; after bringing again the devices to 22 °C, it was observed an irreversible PCE loss of (21.6 ± 2.3) %. From the *I*-*V* curve of a representative device, Figure 3, when the cell returns to room temperature after 80 °C, it is observed a decrease in the J_{sc} and V_{oc} due to enhanced series and shunt resistances.

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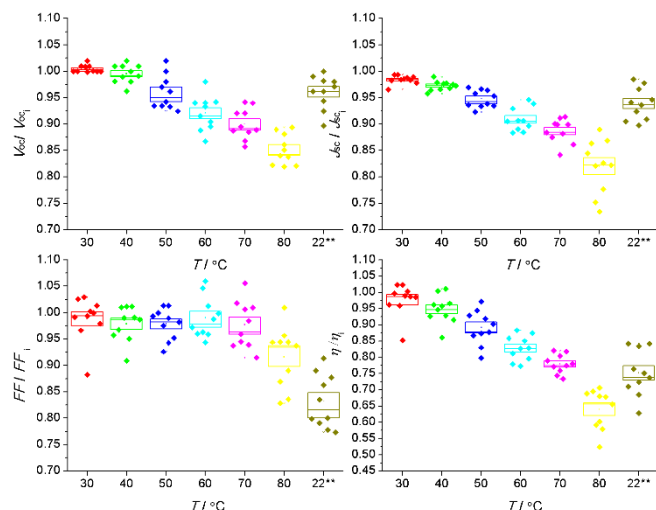


Figure 2. Photovoltaic performance of complete PSC devices at temperatures between 30 °C and 80 °C. The photovoltaic performance values were normalized by the initial PCE obtained at 22 °C for each cell. It was also added the normalized performance parameters of cells characterized at 22 °C after high-temperature stress tests (22**).

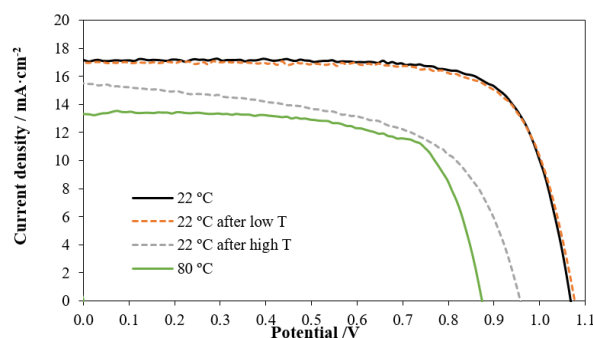


Figure 3. *I*-*V* curves of a representative device with spiro-MeOTAD as HEL operating at room temperature (22 °C) (black line) and the same cell operating at 80 °C (green line). Control characterization were performed at 22 °C after thermal-stress at low (orange dotted line) and high temperatures (grey dotted line).

The PCE loss observed at high operating temperatures can be assigned either to spiro-MeOTAD or perovskite degradation. Scanning electron microscopy images of cells exposed to 80 °C showed pinholes on top of the HEL layer – Figure 4. According to Jena *et al.*,^[32] pinholes only appear when gold is on top of spiro-MeOTAD and when Li-TFSI and TBP are used together as dopants of HEL. Interestingly, Figure 4 shows pinholes over a spiro-MeOTAD surface region of the PSC without gold back contact; pinholes are formed after thermal-stress test and independently of this surface area being covered or not by a gold back-contact layer.

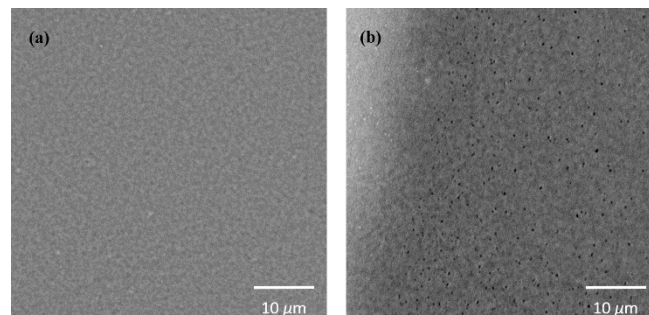


Figure 4. Top SEM micrographs of a mesoporous triple-cation perovskite solar cell in an area without gold back contact on top (a) before and (b) after thermal-stress tests.

For understanding the loss of PCE of a PSC under thermal-stress, two set of experiments were conducted. First, the gold back-contact layer was peeled-off from a batch of fresh and a batch of thermally aged cells (80 °C for 4 h in a hot plate) and a new layer of gold was evaporated on top of spiro-MeOTAD. These devices were characterized at room temperature, however the degradation remained in the batch of thermally aged cells. Then, the gold back-contact and spiro-MeOTAD layers were removed from new batches of fresh and thermally aged cells and fresh layers of gold and spiro-MeOTAD were deposited – Table 1. Again, these devices were characterized at room temperature and the PCE recovered only after the use of a fresh layer of spiro-MeOTAD. This indicated that the loss of performance after the thermal-stress is related to HEL layer probably due to the loss of volatiles, either additives or degradation products.

Table 1. Photovoltaic parameters obtained at 22 °C of a PSC device thermally-aged at 80 °C and after the removal and deposition of a new layer of spiro-MeOTAD and gold.

Photovoltaic parameters	Fresh cell	Thermally-aged cell	Cell with new layers
V_{oc} [mV]	945	924	964
J_{sc} [mA·cm ⁻²]	17.0	16.6	17.3
<i>FF</i>	0.644	0.558	0.649
η [%]	11.4	9.40	11.9

To corroborate the effect of additives loss in the photovoltaic performance, spiro-MeOTAD without additives was used to assemble a new batch of cells. In this case the power conversion efficiency increased with temperature, Figure 5, that was attributed to the heating-induced improvement of the crystallinity, promoting a more efficient charge mobility and correspondent extraction.^[20] Since the glass transition temperature of the spiro-MeOTAD is 125 °C^[33] and there is no degradation in pristine (not additivated) spiro-MeOTAD, it can be conclude that the performance decrease should be related to evaporation of additives. This is further supported since the performance of spiro-MeOTAD-based PSC without additives is similar to the performance of additivated spiro-MeOTAD-based PSC after thermal-stress – Table 2. The pinholes at the spiro-MeOTAD layer should then be formed because the fast evaporation of additives

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during the thermal-stress tests. It should be emphasized that due to the pressure rise inside the cell, it should leak out during the evaporation of the low boiling point solvents. Even though TBP presents a boiling point of almost 200 °C, Bailie *et al.* showed that this compound in small quantities can possess an appreciable equilibrium partial pressure that allows it to evaporate at temperatures as low as 85 °C in dye-sensitized solar cells.^[34] Finally, and as discussed before, also Kasparavicius *et al.*^[25] showed that oxidized HEL starts to degrade at those values of temperature and part reverts back to the original unoxidized material, and other part reacts with TBP.

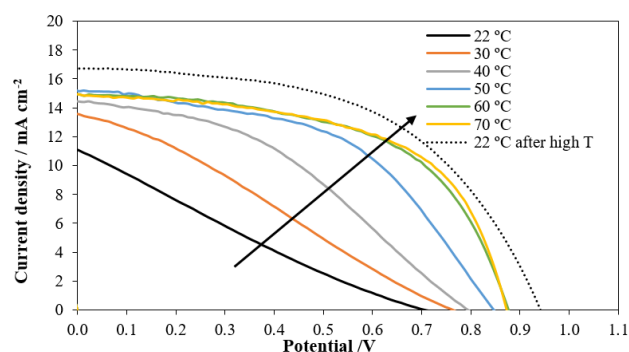


Figure 5. *I-V* curves of a representative device, prepared with pristine spiro-MeOTAD, operating at different temperatures, during the thermal-stress tests.

Table 2. Photovoltaic parameters of spiro-MeOTAD cells as HEL, with and without additives at 22 °C after thermal-stress test at 80 °C.

Photovoltaic parameters	Additivated spiro-MeOTAD (TBP + Li-TFSI + FK209)	Pristine Spiro-MeOTAD
V_{oc} [mV]	940	955
J_{sc} [$\text{mA}\cdot\text{cm}^{-2}$]	16.5	15.5
FF	0.551	0.583
η [%]	9.64	9.61

XPS analysis of pristine and additivated spiro-MeOTAD before and after thermal-stress tests were performed. This analysis allows to assess possible migration to the surface of additives species when additivated spiro-MeOTAD is thermally aged – Figure 6.

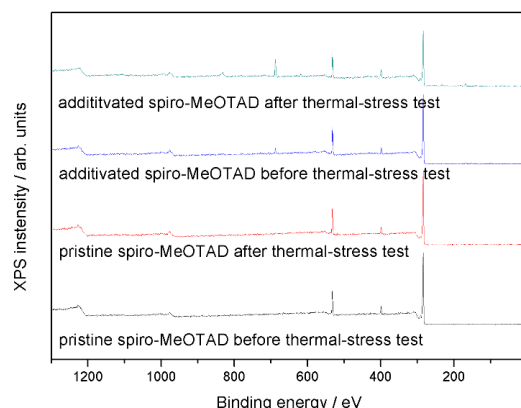


Figure 6. XPS survey spectra (Al K α = 1486.6 eV) of pristine and additivated spiro-MeOTAD before and after thermal-stress tests.

The spectrum of pristine spiro-MeOTAD before thermal stress shows C 1s, O 1s and N 1s XPS peaks similar to those of the pristine spiro-MeOTAD spectrum after thermal stress. This means that no significant changes occur in the non-additivated HEL during tests. When comparing fresh pristine and additivated spiro-MeOTAD, it is observed a small peak at ~688 eV in this sample, corresponding to the F 1s peak, attributed to the $-\text{CF}_3$ groups of Li-TFSI and FK209-TFSI additives. On the other hand, lithium, sulfur and cobalt (from FK209) should also display a detectable signature. Figure 7 zooms the spectra of the additivated sample, before and after thermal stress, for assessing Li 1s, Co 2p, F 1s and S 2p. The overlapping signals of C-H, C-C and C-N from spiro-MeOTAD and TBP turns difficult the detection of TBP additive in the XPS spectrum.^[35]

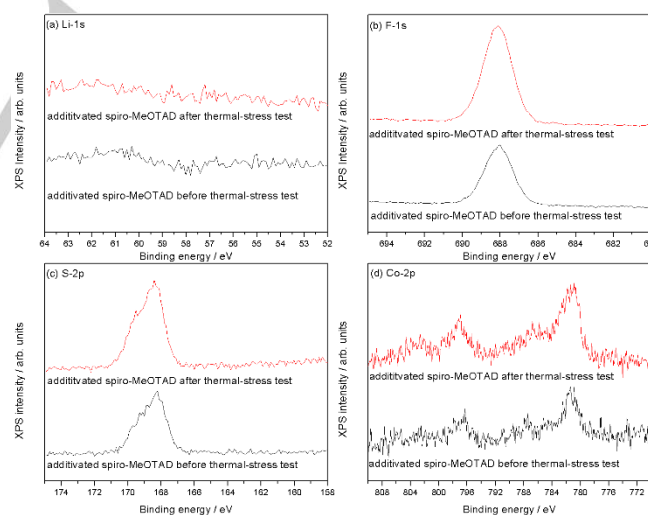


Figure 7. XPS spectra (Al K α = 1486.6 eV) corresponding to (a) Li-1s, (b) F-1s, (c) S-2p and (d) Co-2p core levels of the spiro-MeOTAD films at room temperature before and after thermal-stress tests.

Changes in Li 1s region are undetectable because of low response sensitivity factor of lithium (RSF = 0.025) compared to the other elements (1 to F 1s, 0.668 to S 2p and 3.59 to Co 2p). In additivated spiro-MeOTAD, clear XPS signals of F 1s, S 2p and Co 2p are observed and their intensity increases after thermal

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stress. In particular, the atomic percentage of F increases from 6.5 to 10.9 at. %; this increase evidences a migration phenomenon of these elements to the surface of spiro-MeOTAD, promoted during the evaporation of volatiles during the thermal-stress tests.

Figure 8 shows the characteristic curves of a fresh cell using PTAA as hole extraction layer at room temperature (22 °C), under thermal stress at 80 °C, 70 °C, 60 °C and 50 °C, and back to room temperature. It can be seen that with PTAA the cell recovers mostly its original performance. Actually, this explains why PTAA has been reported to produce very stable cells with minimal PCE losses, even when running at 85 °C for 500 h, since these aged devices are then characterized at room temperature.^[2] The photovoltaic performance as a function of the temperature shows no significant impact in J_{SC} for PTAA device but a strong decrease in the FF and V_{OC} is observed, as a consequence of an increase of series resistance. Khadka *et al.*^[36] reported similar behavior mentioning that this can happen because a number of factors such as accelerated interfacial recombination, bulk defect activities or contact limited transport. When devices are submitted to high temperatures and then returned back to the room temperature, the additives evaporate and so the mobility of these aged HELs becomes closer to the pristine values. In fact, hole mobility of pristine PTAA ($\approx 1 \times 10^{-2} - 1 \times 10^{-3} \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$)^[37] is higher than pristine spiro-MeOTAD ($1.6 \times 10^{-5} \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$)^[38] and so it is understandable that the irreversible loss is higher for the spiro-MeOTAD-based devices.

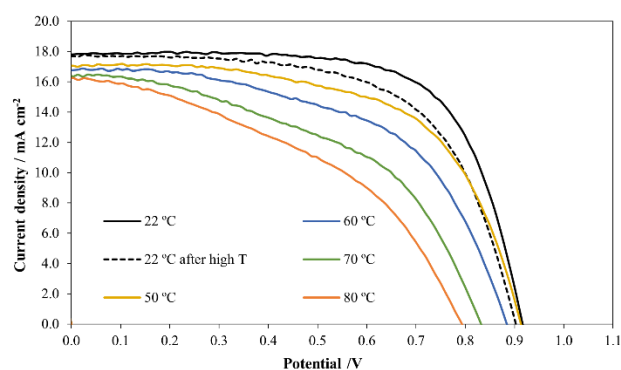


Figure 8. I - V curves of a representative device, with PTAA as HEL, operating at room temperature (22 °C) in the beginning of the test (black line) and after working at 80 °C (dashed line); and operating at 80 °C, 70 °C, 60 °C and 50 °C (colored lines).

PTAA-based cells displayed an irreversible loss of PCE after thermal-stress of $(8.2 \pm 1.6) \%$ which compares very favorably with the loss of $(21.6 \pm 2.3) \%$ observed for the spiro-MeOTAD-based cells – Table 3. Interestingly, the PTAA-based cells display a higher decrease in PCE under 80 °C compared with spiro-MeOTAD-based cells.

Table 3. Loss and recovery of power conversion efficiency in different HEL tested in thermal stress-studies.

HEL	Decrease at 80 °C [%] - average	Irreversible loss ^[a] [%] - average
spiro-MeOTAD	36.0 ± 5.5	21.6 ± 2.3
PTAA	48.2 ± 9.3	8.2 ± 1.6

[a] values at 22 °C, after high-temperature thermal-stress tests.

Similarly to spiro-MeOTAD-based cells, pinholes were also visible in thermally stressed PTAA-based cells – Figure 9. PTAA also contains TBP and Li-TFSI additives, so they will also volatilize with temperature. However, this may occur in less extension since additives are added to PTAA at lower concentration.

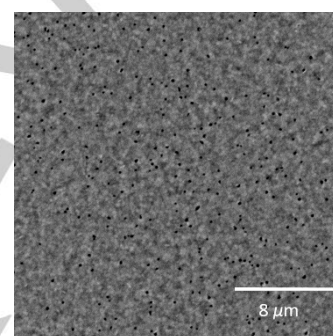


Figure 9. Top SEM micrographs of a mesoporous triple-cation perovskite solar cell with PTAA as HEL after thermal-stress tests.

After assessing the role of the HEL additives loss on the photovoltaic performance, it is now important to assess the role of the perovskite after the thermal-stress test and a possible degradation. According to some authors,^[39,40] the presence of additives and moisture exacerbates the degradation of perovskite absorber when the layer is in direct contact with spiro-MeOTAD. XRD diffractogram of devices with spiro-MeOTAD and PTAA after thermal-stress did not show any differences compared with XRD diffractogram at room temperature performed before encapsulation – Figure 10. Several devices were analyzed and none of them displayed PbI_2 peak, corresponding to (0 0 2) diffraction peak, common sign associated to the degradation of the perovskite crystal. These results indicate that there is no degradation on perovskite layer.

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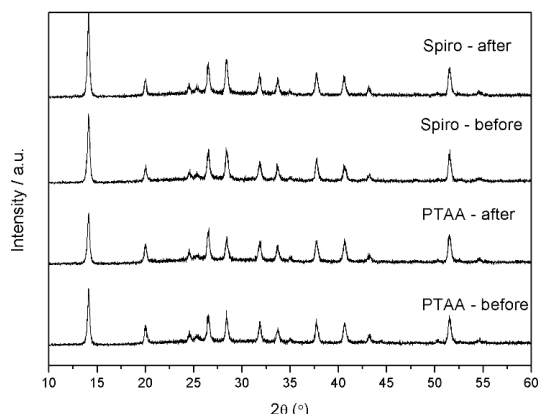


Figure 10. XRD diffractogram of triple cation perovskite devices using PTAA and spiro-MeOTAD as HEL, before and after thermal-stress tests at 80 °C.

Conclusions

Temperature influence is critical for perovskite solar cells since during real conditions of outdoor operation the device can easily reach temperatures 45 °C higher than room temperature. Within this work PSC devices are exposed to thermal-stress in a range of temperatures between -5 °C and 80 °C to reproduce real operating conditions, allowing to investigate the temperature effect on the devices performance. The performance of mesoporous triple cation perovskite cells displayed ca. 5 % less PCE at -5 °C when compared to 22 °C, which indicates a small sensitivity of this technology to low temperatures; the thermally-stressed samples were then characterized at 22 °C, displaying essentially the same photovoltaic performance. However, at 80 °C it was observed a significant decrease in the V_{oc} and J_{sc} , leading to an average PCE decrease of $(36.0 \pm 5.5) \%$. Also, this thermally-stressed samples when characterized at 22 °C displayed an irreversible performance loss of $(21.6 \pm 2.3) \%$. The roots of this performance loss were investigated. It was concluded that temperatures between -5 °C and 80 °C do not affect the perovskite layer since no peak of PbI_2 was observed in the XRD diffractogram; instead, the high-temperature stress test originates the evaporation of the additives presented in the HEL that is responsible for the irreversible performance loss. This conclusion was supported by the appearance of pinholes in the spiro-MeOTAD layer, observed by SEM, and by the percentage increase of elements like F, S and Co (characteristic XPS peaks of Li-TFSI and FK209 compounds) on the top surface of spiro-MeOTAD after thermal-stress test, observed by XPS. The migration of elements derived from volatiles evaporation. Also, it was observed at 80 °C that a PSC prepared with a spiro-MeOTAD layer without additives displays the same photovoltaic performance of a device prepared with additivated spiro-MeOTAD. Besides spiro-MeOTAD, PTAA is often used as HEL in PSC. The PTAA needs less additives when compared with spiro-MeOTAD, indicating a potentially better thermal stability. Actually, it was concluded that PTAA-based devices displayed an irreversible performance loss of only $(8.2 \pm 1.6) \%$. This reinforces the role of the additives on the PSC performance loss upon thermal-stress.

Therefore, more efforts need to be done to devise thermally stable and highly conductive HELs.

Experimental Section

Materials and device fabrication: Triple cation perovskites prepared by anti-solvent method were used to prepare complete devices as described elsewhere.^[3] Fluorine-doped tin oxide (FTO) glass substrates (TCO 22-7, 7 Ω /square, Solaronix, Switzerland) were cleaned with KOH saturated solution and water in an ultrasonic bath, 5 min each step. The substrates were then dried with air and treated with UV-ozone for 15 min (UVO cleaner from Jelight Company, Inc.). At 5 mm from the glass edge, a scribing was made with a Versa Laser printer to electrically isolate the photo- from the counter-electrode.

Titanium dioxide (TiO_2) blocking layer was made from a precursor solution of titanium diisopropoxide bis(acetylacetonate) in anhydrous ethanol sprayed at 450 °C and left at that temperature for 45 min. After cooling down to room temperature, TiO_2 paste (Dyesol, 30 NR-D) was spin-coated on the blocking layer at 5000 rpm for 10 s and immediately placed at 125 °C. After drying at 125 °C for 10 min, the film was annealed at 500 °C for 30 min.

The triple cation perovskite solution was prepared from a precursor solution containing formamidinium iodide (FAI - 1 M), lead iodide (PbI_2 - 1.1 M), methylammonium bromide (MABr - 0.2 M) and lead bromide ($PbBr_2$ - 0.2 M) in anhydrous N,N-dimethylformamide:dimethyl sulfoxide (DMF:DMSO 4:1 (v:v)). After the total dissolution of all the components, 5 % in volume of a 1.5 M cesium iodide (CsI) solution in DMSO was added to the initial solution. The perovskite layer was deposited by spin-coating at 1000 rpm during 10 s, followed by 30 s at 6000 rpm; 15 s before the end of second step, 100 μ L of chlorobenzene was deposited on top of the spinning substrate and immediately a brownish color appeared. The substrates were placed at 100 °C during 40 min to anneal. After the annealing, the substrates were cooled down for few minutes before the deposition of HEL.

The HEL solution was prepared dissolving 102 mg of spiro-MeOTAD (Chemborun, 99.57 %) in 1.116 mL of chlorobenzene, to which 40 μ L of 4-*tert*-butyl pyridine (TBP - Acros Organic, 96 %), 23 μ L of lithium bis(trifluoromethanesulfonyl)imide (Li-TFSI - Acros Organic, 99 %) solution (520 mg Li-TFSI in 1 mL acetonitrile) and 10 μ L of FK209 (Dynameo) solution (40 μ g in 100 μ L of acetonitrile), were added. 40 μ L of this spiro-MeOTAD solution was spin-coated on the perovskite layer at 4000 rpm for 20 s. To the samples with PTAA [poly[bis(4-phenyl)(2,4,6-trimethylphenyl)amine] as HEL, the solution was prepared dissolving 10 mg of PTAA (Ossila) in 1 mL of toluene and added 2.0 μ L of TBP and 1.6 μ L of 1.8 M of Li-TFSI solution. Finally, ca. 60 nm gold electrode was deposited by thermal-evaporation on the spiro-MeOTAD layer. The perovskite and spiro-MeOTAD/PTAA depositions were performed inside a nitrogen filled glove box.

Complete devices were sealed using double side Kapton® tape and the cell edges coated with high temperature epoxy resin (Original Cold-Weld® two-part epoxy steel reinforced from JB Weld®; product number: 8265-S). No effect of Kapton® and epoxy resin on devices photovoltaic performance was observed. It is important mentioning that the pressure inside the cell increases as the temperature increase. Not only the permanent gases contribute for the pressure rise but mostly the evaporation of low boiling solvents, which make the cell eventually to leak. This leak should, however, not impair the performance of cell since the amount of air entering in the cell during the experimental time is rather limited.

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Characterization and experimental setup: Photocurrent – potential (*I*-*V*) characteristics curves were obtained in a setup equipped with a 150 W xenon light source (Oriol class A simulator, Newport USA) irradiating ca. 98 ± 2 mW·cm⁻² with an air mass filter of 1.5 (Newport, USA). The *I*-*V* curves were obtained by applying an external potential bias and measuring the generated photocurrent with an Autolab electrochemical station in the temperature range of -5 °C to 80 °C. The measurements were done in reverse scan (from open-circuit to short-circuit) at scan rate of 10 mV·s⁻¹ and after stabilization of the *V*_{oc}. This stabilization was fast (less than 20 s) to ensure that the characterization is not influenced by the light-soaking effect.

The temperature of the cell was controlled using an in-house developed setup based on a peltier element (Marlow Industries, model RC12-6) connected to a Keithley DC power supply (Model 2425C) and controlled by a National Instruments LabVIEW software.^[41] Three K-type thermocouples were used to control the temperature: one in the cell, to read the real operation temperature, and the other two at the different sides (cold and hot sides) of peltier device – cf. Figure 11. The heat dissipation side of peltier element was connected to a thermostatic water bath (Julabo model ME, Germany). The devices were submitted to 12 different temperatures, from -5 °C to 80 °C, in a step-wise process with a dwell time at each temperature of 10 min; at end of each dwell time the *I*-*V* was recorded. The overall experiment lasted ca. 4 h for each cell.

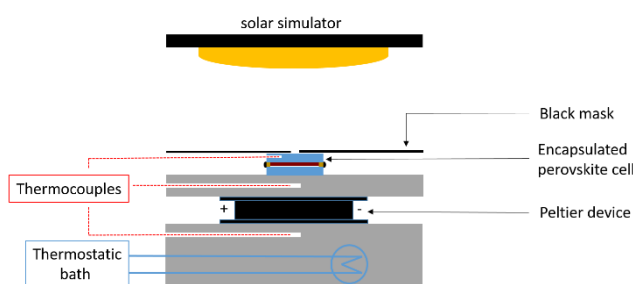


Figure 11. Experimental setup used for PSC temperature control between -5 °C and 80 °C; grey components are aluminum slabs.

The XPS analysis was performed in a Kratos Axis Ultra HAS spectrometer (from Kratos Analytical, England) using monochromatic Al K α radiation (1486.6 eV), operating at 15 kV (300 W). The surface atomic percentages were calculated from the corresponding peak areas using the sensitivity factors provided by the manufacturer. The top morphology of HEL was analyzed by SEM in a Phenom XL desktop scanning electron microscope from PhenomWorld. The X-ray diffraction patterns of the devices were obtained in a Rigaku SmartLab diffractometer with Cu K α radiation in the parallel beam geometry.

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Keywords: energy conversion • hole transport material • perovskite solar cells • photovoltaics • temperature degradation

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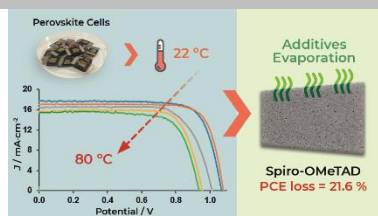
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Entry for the Table of Contents (Please choose one layout)

Layout 1:

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Temperature has a great impact in the perovskite solar cells performance during their operation. The increase of temperature originates the evaporation of the additives presented in the hole extraction layer. The additives evaporation is reflected in an irreversible loss of 21.6 % of power conversion efficiency using spiro-OMeTAD and 8.2 % using PTAA.



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